ECEN-662: Homework 6

Due on March 25, 2026 at 11:59 PM

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Problem 1

Consider a Gaussian Mixture Model (GMM) with two well-separated modes:

$$p(\theta) = 0.5 \cdot \mathcal{N}(\theta; -3, 1) + 0.5 \cdot \mathcal{N}(\theta; 3, 1),$$

where $\mathcal{N}(\theta; \mu, \sigma^2)$ is a Gaussian density with mean μ and variance σ^2. Our goal is to approximate the above GMM using the variational distribution:

$$q_\lambda(\theta) = \mathcal{N}(\theta; \mu, \sigma^2),$$

where the variational parameter $\lambda = (\mu, \sigma)$.

i) Derive the explicit update rule for both μ and σ using the re-parameterization trick.

ii) Write a computer program to find the optimal variational distribution for the above GMM model. Try multiple initial values for μ and σ and describe how the value of μ and σ change over the iterations.

iii) Repeat Part ii) for the following updated GMM model:

$$p(\theta) = 0.5 \cdot \mathcal{N}(\theta; -3, 4) + 0.5 \cdot \mathcal{N}(\theta; 3, 0.25).$$

Solution